AGENTIC AI • HACKATHON PROJECT

CandiQ

Evidence-backed recruitment intelligence

Discover candidates. Verify claims. Explain every recommendation.

PYTHON • GROQ • CRUSTDATA

Team CandiQ | ContextCon IIT Bombay

Recruitment research is fragmented and manual

Finding candidates is only the first step. Evaluating evidence is where the work compounds.

01 Search overload

Recruiters search many profiles and job-title variations. Relevant candidates can be missed by narrow queries.

02 Inconsistent screening

Profile details are unstructured. Manual comparisons can lead to uneven assessments.

03 Unsupported assumptions

A profile mention is not always proof of a specific skill, tool, or level of experience.

The opportunity: automate repetitive sourcing and make the reasoning behind a shortlist inspectable.

From a hiring brief to an evidence-backed shortlist

A recruitment research assistant — not an autonomous hiring decision-maker.

RECRUITER INPUT

“Find a senior ASIC Physical Design Engineer in Bengaluru...”

CANDIQ WORKFLOW

01 Understand requirements

02 Discover & enrich profiles

03 Extract and verify evidence

04 Score with transparent rules

RECRUITER OUTPUT

Ranked shortlist
Evidence links
Missing information
Review flags

Human recruiters retain control over final assessments and hiring decisions.

A modular, multi-stage agentic workflow

LLM reasoning is combined with external data retrieval and deterministic Python logic.

01	02	03	04	05	06	07
Recruiter brief	**Requirement parser**	**Candidate search**	**Pre-rank + enrich**	**Claim extraction**	**Evidence check**	**Score + report**
Natural language	GPT-OSS-20B	Crustdata API	Python + API	LLM	LLM + validation	Python

OUTPUT → Evidence-backed shortlist for human recruiter review

Structure first. Search second.

The system translates an open-ended brief into explicit, prioritized criteria.

Requirement understanding

- Role, seniority and location
- Must-have vs nice-to-have
- Search variants and assumptions
- Structured output validated with Pydantic

Candidate discovery

- Crustdata search filters
- Strict and relaxed variants
- Pagination and deduplication
- Candidate pre-ranking

Selective enrichment

- Enrich selected profiles
- Batch requests and throttling
- Retry handling
- Local response caching

Design principle: retrieve broadly enough to avoid narrow-title misses, then enrich selectively.

A recommendation must be traceable to source data

CandiQ evaluates atomic claims rather than trusting a single generated summary.

SOURCE PROFILE

Title: Senior ASIC Physical Design Engineer

Experience: Timing closure

Tools: Synopsys PrimeTime

ATOMIC CLAIMS

- ASIC physical design
- Timing analysis
- Synopsys tool experience

EVIDENCE CHECK

Each claim is linked to a source record, field path and value, then verified.

Illustrative example: every claim must be checked in its actual source context.

Missing evidence is not proof of missing skill

Four verification outcomes prevent unsupported claims from silently becoming facts.

SUPPORTED	Direct evidence supports the claim
PARTIAL	Evidence is incomplete
UNSUPPORTED	No adequate evidence found
CONTRADICTED	Evidence conflicts with the claim

Example: PrimeTime supports Synopsys + timing; STA alone does not prove a vendor.

Separate candidate fit from evidence quality

Weighted Python scoring makes the evaluation inspectable and repeatable.

WEIGHTED CRITERIA

Must-have	Weight 3
Nice-to-have	Weight 1
Yes	100% credit
Partial	50% credit
No / Unknown	0% credit

TWO DISTINCT SIGNALS

Candidate score

How well assessable criteria are met

Evidence coverage

How much of the evaluation is supported

Low evidence coverage or unknown must-have criteria can trigger human review.

A practical stack with clear responsibility boundaries

LLM reasoning where language is needed; deterministic code where consistency matters.

LLM	**GPT-OSS-20B via Groq**	Requirement parsing, claim generation and verification
DATA	**Crustdata API**	Professional profile search and enrichment
BACKEND	**Python**	Orchestration, ranking, scoring and reporting
VALIDATION	**Pydantic + tests**	Schemas, evidence-path checks and regression coverage

A working end-to-end pipeline

Illustrative results from one recorded live run — not a performance guarantee.

5	5	28	6	23
profiles retrieved	profiles enriched	claims generated	claims rejected	automated tests passed

The value is not just a shortlist — it is the evidence trail behind it.

Built with clear boundaries — designed to grow

We distinguish what is implemented today from the next product milestones.

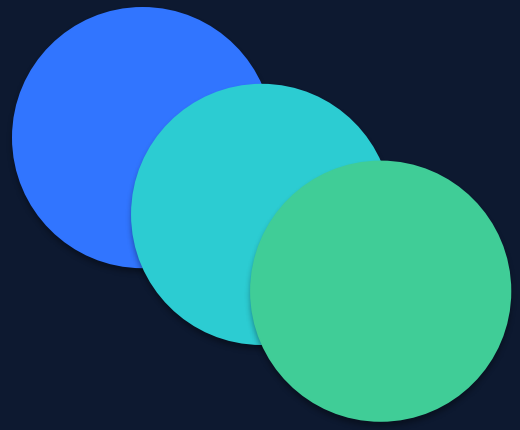
Current limitations

- Data quality depends on source completeness and freshness
- Search filters have API-specific constraints
- LLM outputs require validation
- Limited production scalability
- No full recruiter dashboard yet

Future roadmap

- Specialized agents for sourcing and verification
- Semantic + hybrid candidate retrieval
- Cross-source verification and confidence calibration
- Recruiter dashboard and configurable criteria
- Scalable processing and fairness monitoring

Don't just recommend.
Show the evidence.



Discover candidates. Verify claims. Support better-informed human decisions.

THANK YOU • Q&A